Faculty of Engineering, Ain Shams University
Credit Hours Engineering Program
Mechatronics & Automation Program

CSE 411

Real Time and Embedded Systems

ON/OFF Temp Control Project

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Abstract

Our project functionality is to control the temperature using a heater and receiving feedback from a potentiometer in order to maintain the setpoint by using queues to send data between tasks and communicating with the user via a terminal to change the setpoint temperature and display the setpoint and measured temperature on the LCD. The heater will switch on if the temperature falls below the setpoint, and it will turn off if the temperature rises over it.

Inputs and Outputs

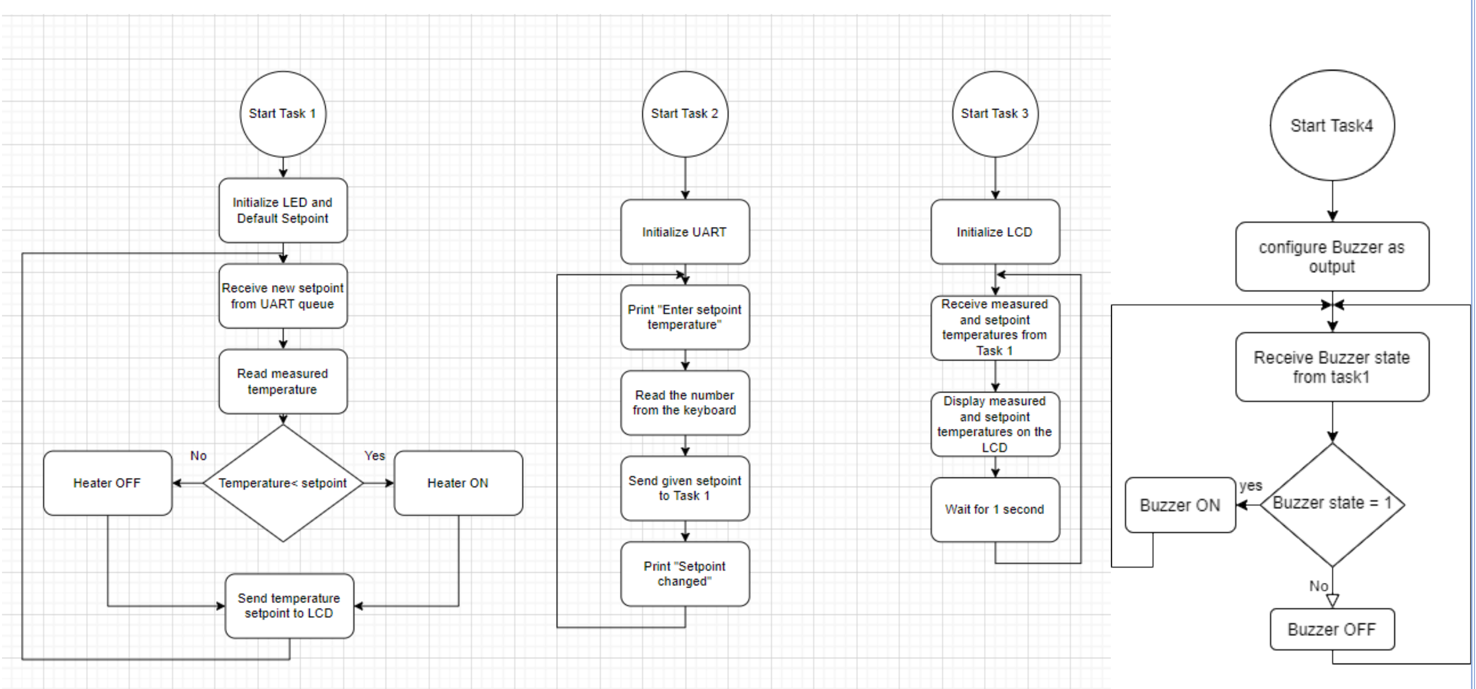
Inputs

- Potentiometer
- User's setpoint temperature

Outputs

- LED on PF3
- Heater on PF2
- Buzzer on PF1

Flowchart



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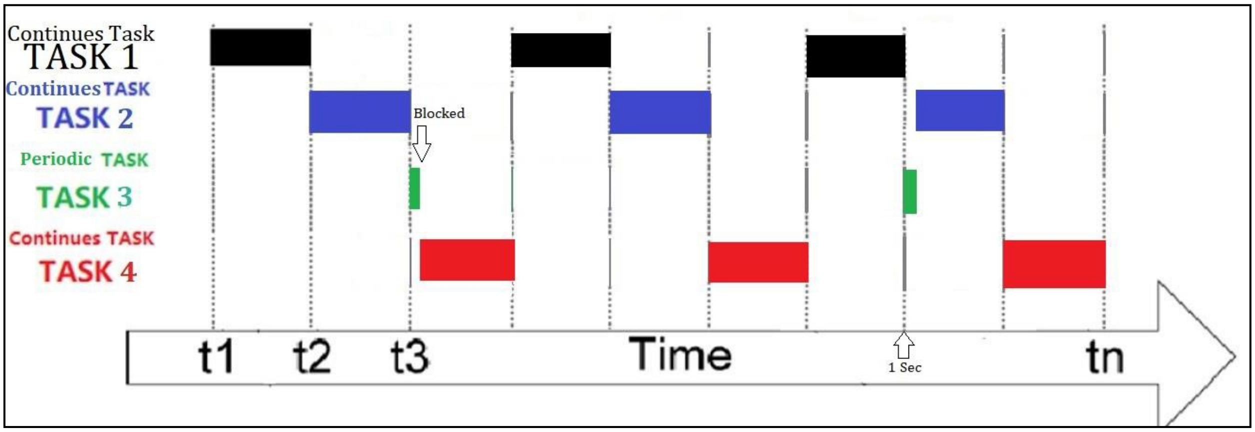
Task Diagram

All tasks have the same priority



Description of Files and Tasks

Files

Main.c : Main file including the 4 tasks, initialization function main function. LCD_Config.c: A Library file for implementing LCD functions.

ADC.c: A Library file for implementing ADC functions.

Functions

Task1: Main Controller function for comparing temperature with setpoint and set on or off the heater through the queue send and receive.

Task2: UART Controller function for displaying the communication between System and laptop and changing the setpoint temperature and send the new setpoint to task1 through UART queue.

Task3: LCD Controller Function for displaying on LCD the measured temperature from the potentiometer and the setpoint by receiving the data from task1 through UART.

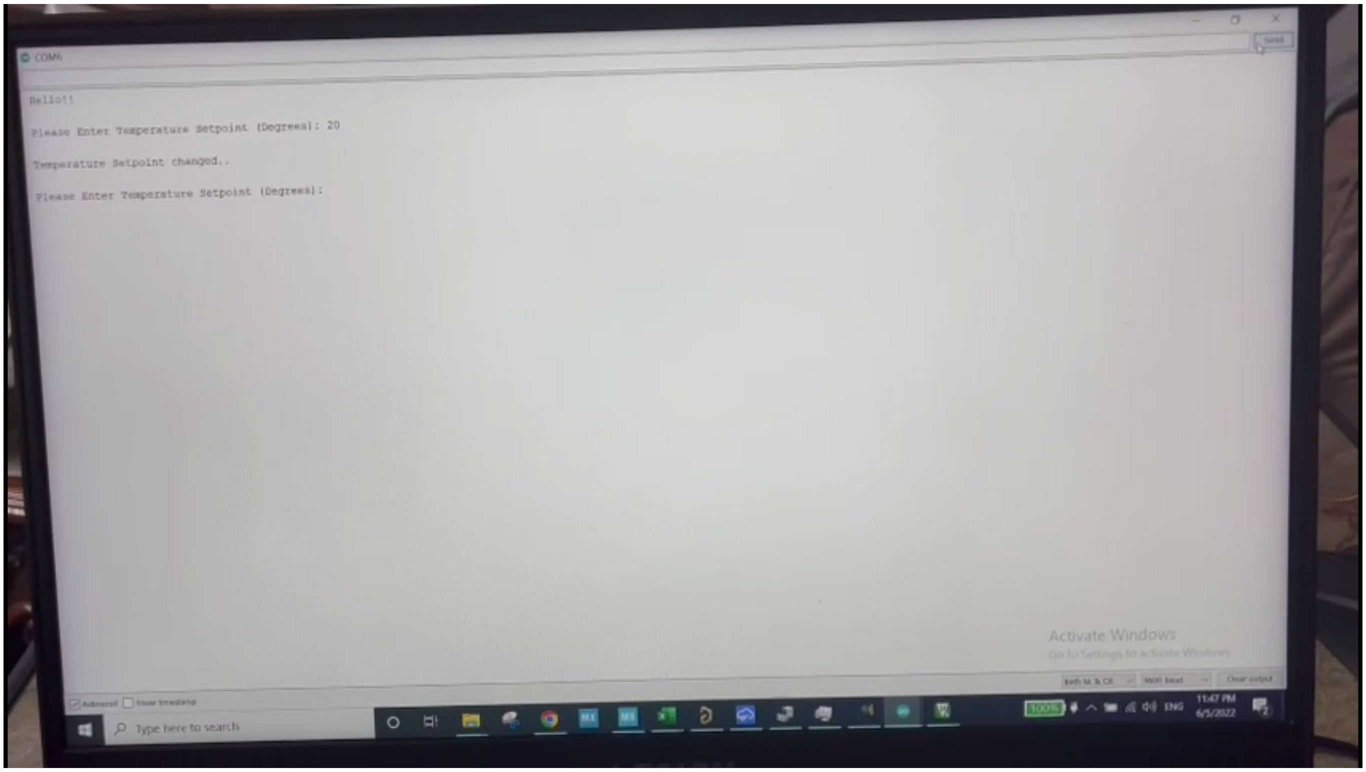
Task4: Buzzer controller function to check if task1 sent on the queue data to set on the buzzer ornot by receiving data from task1 through Buzzer Queue.

GPIO_init: An initialization function to initialize GPIO ports and UART.

Main: main function to create queues and tasks for the program and start the scheduler

Outputs

Serial Monitor Output



Libraries and Functions

LCD Library

A readymade LCD driver library was used to display potentiometer readings on it. It used the nibble approach as to reduce the number of wire connections required, but unfortunately our LCD had a problem and didn't work with any code or device.

ADC Driver

To use the potentiometer an ADC library was made to read and convert readings from the potentiometer.

- Adc_init: to initialize the ADC and read from the potentiometer

UART Functions

To send characters and strings to the terminal on putty on the laptop.

- UART_Init: UART initialization
- UART0_Receiver: Receives data entered by user
- UART0_Transmitter: Transmits data

For project code and demonstration video click here